

GitHub Commits & Pull Requests

SISMID Lab

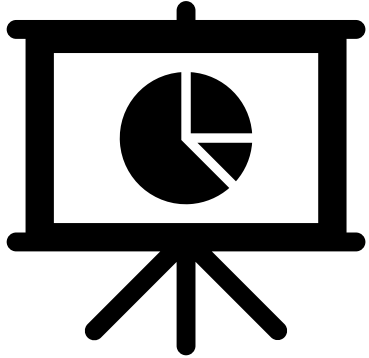
July 22 through July 24, 2026

A Human-AI Teaming Approach to Infectious Disease Modeling

Lecture Overview

Topic	Slides
1. General Housekeeping	3
2. Lab Timeline	4
3. Task Overviews	5
4. Goal Output	6
5. Task 1: GitHub Commits	7–24
Commit via VS Code	9–15
Commit via GitHub	16–24
6. Task 2: Submitting a Pull Request	25–30

General Housekeeping

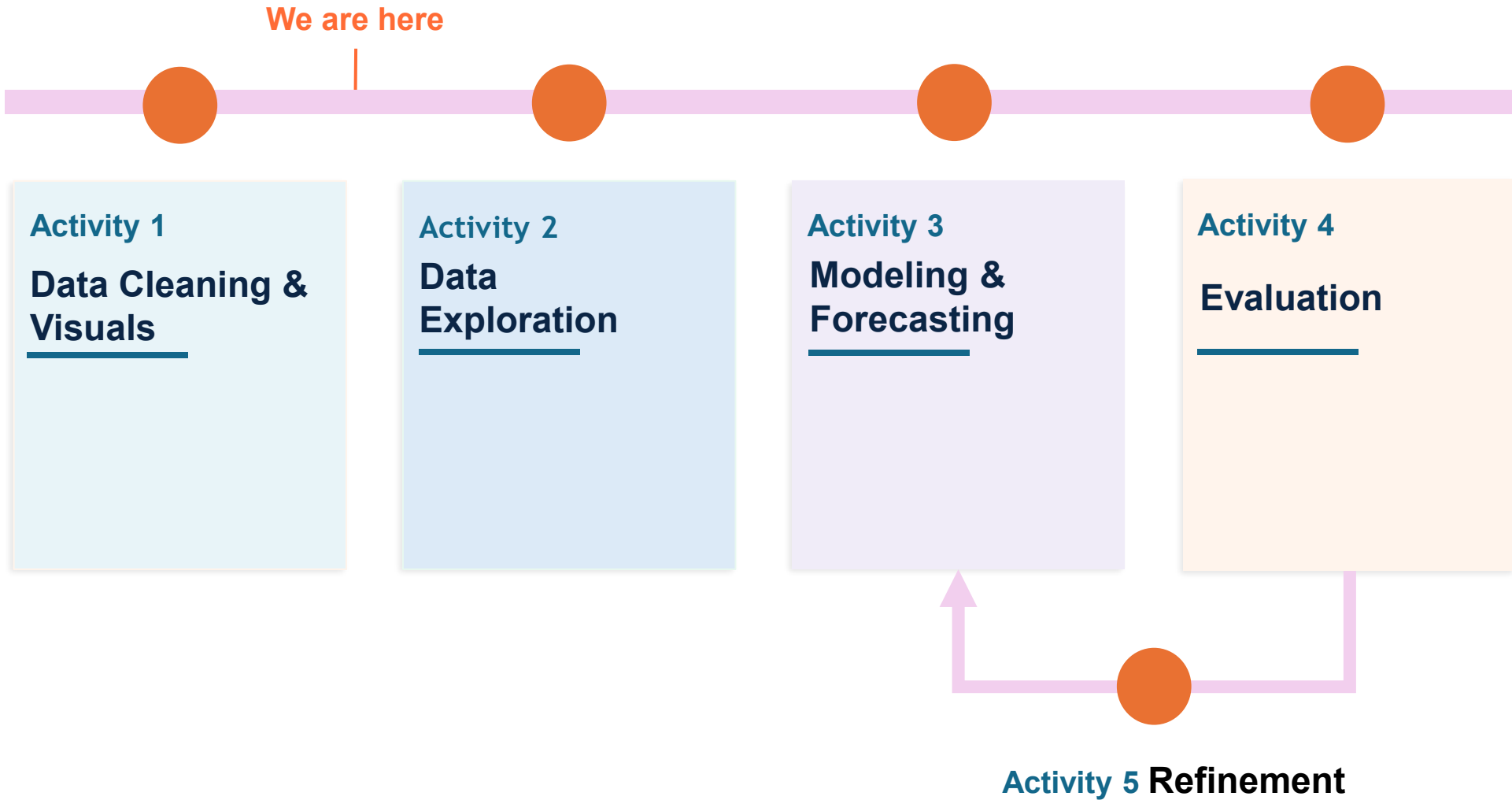


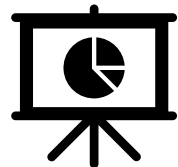
Any slide with this icon contains the **overview** and **expected outputs** for the included tasks. All outputs listed on these slides should be uploaded to the course GitHub repository for review.



Any slide with this icon indicates that it is **time to complete a hands-on activity**. Carefully follow the instructions provided, complete the required task, and verify your results.

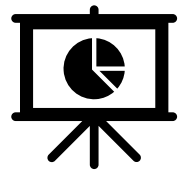
Lab Timeline





Task Overviews

Task	Description
Task 1: GitHub Commits	Upload your output files to your GitHub fork. We cover two ways to do this: through <i>VS Code</i> and through the <i>GitHub web portal</i> .
Task 2: Submit a pull request	Open a <i>pull request</i> to merge the changes from your fork into the main class repository.



Goal Output

After completing this activity, you should have an **updated GitHub fork** with your results committed, and a **pull request** that merges those changes into the main class repository.

Activity Output	Description
Updated GitHub fork	Your output files committed and uploaded to your GitHub fork.
Pull request	A request to merge the changes in your fork into the main class repository.



Task 1: GitHub Commits (Uploading Results)

File Edit Selection View Go Run Terminal Help

01_cleaning.R U X

Lab-Materials > Day 1 - Lab Materials > Activity 1 > START HERE > scripts > 01_cleaning.R > abc ==

```
1 # =====
2 # 01_cleaning.R
3 #
4 # Data Cleaning Agent: clean_flu_admissions
5 #
6 # Turns the raw NHSN HRD influenza CSV into a tidy, three-column dataset
7 # (week, location, value) and produces a US influenza-admissions epicurve.
8 #
9 # Implements the rules defined in agents.md / rules.md.
10 # =====
11
12 library(readr)
13
14 # -----
15 # Paths
16 # -----
17 base_dir <- file.path("Lab-Materials", "Day 1 - Lab Materials", "Activity 1", "START HERE")
18 input_csv <- file.path(base_dir, "data",
19   |-----| "Weekly Hospital Respiratory Data (HRD) Metrics by Jurisd
20 output_data_dir <- file.path(base_dir, "output", "data", "01_cleaning")
21 output_fig_dir <- file.path(base_dir, "output", "figures", "01_cleaning")
22 output_csv <- file.path(output_data_dir, "cleaned_flu_admissions.csv")
23 output_fig <- file.path(output_fig_dir, "epicurve_us_flu_admissions.png")
24
25 # Create output directories if they do not exist
26 dir.create(output_data_dir, recursive = TRUE, showWarnings = FALSE)
27 dir.create(output_fig_dir, recursive = TRUE, showWarnings = FALSE)
28
29 # -----
30 # Step 3 (helper): Detect the target influenza admissions column
31 # -----
32 allowed_flu_cols <- c("Total Influenza.Admissions", "Total Influenza Admissions")
33
```

EXPLORER

- SISIMID-2026
 - Lab-Materials \ Day 1 - Lab Material...
 - Hints and Answers
 - R-Script
 - VS-CODE
 - START HERE
 - data
 - output
 - data
 - figures
 - scripts
 - agents.md
 - rules.md
 - 01_cleaning.pptx
 - pushing_to_github.pptx
 - output
 - README.md

PROBLEMS 7 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
> # -----
> # Step 8 (cont.): Confirm output files exist
> # ----- .... [TRUNCATED]

> if (!file.exists(output_fig)) {
+   stop("Validation failed: epicurve figure was not created at ", output_fig, call. = FALSE)
+ }

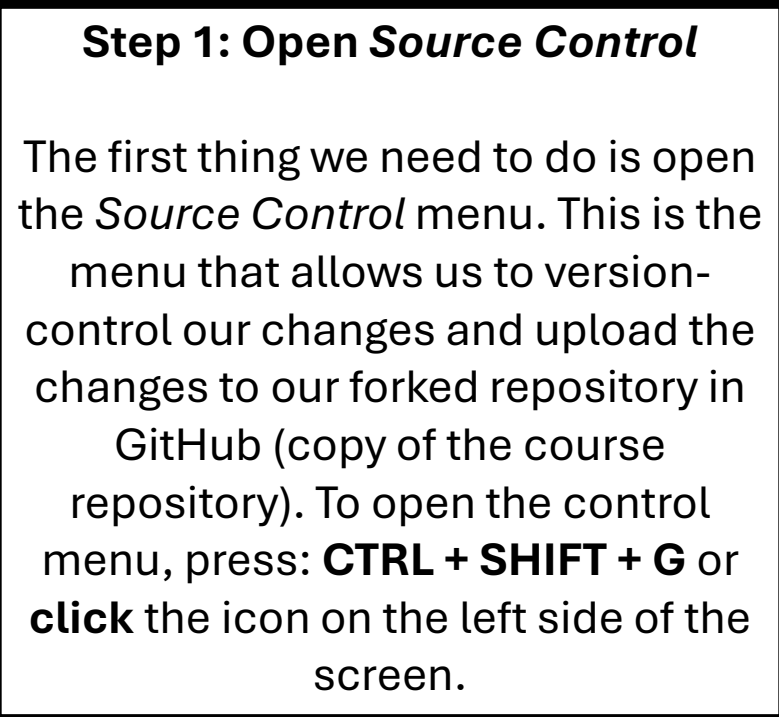
> message("Cleaning complete. ", nrow(cleaned), " rows written to ", output_csv)
Cleaning complete. 307 rows written to Lab-Materials/Day 1 - Lab Materials/Activity 1/START HERE/output/data/01_cleaning/cleaned_flu_admissions.csv

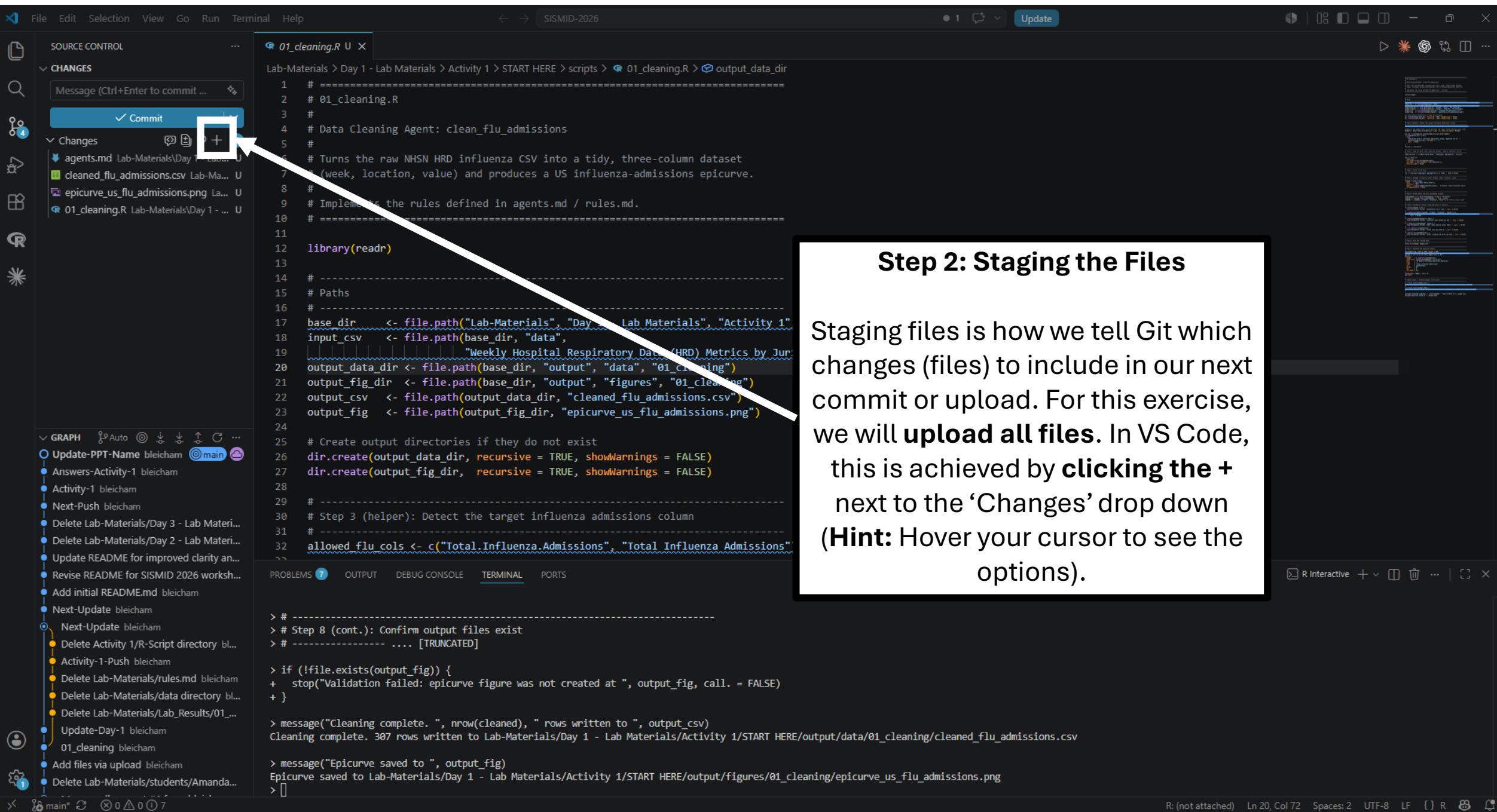
> message("Epicurve saved to ", output_fig)
Epicurve saved to Lab-Materials/Day 1 - Lab Materials/Activity 1/START HERE/output/figures/01_cleaning/epicurve_us_flu_admissions.png
> []
```

After completing Activity 1, you should now see an output folder, along with various figures and data files. We are going to walk through how to “push” or upload them to the course GitHub repository and your copy (fork). We will cover two ways of pushing to GitHub.

R: (not attached) Ln 1, Col 1 Spaces: 2 UTF-8 LF {} R

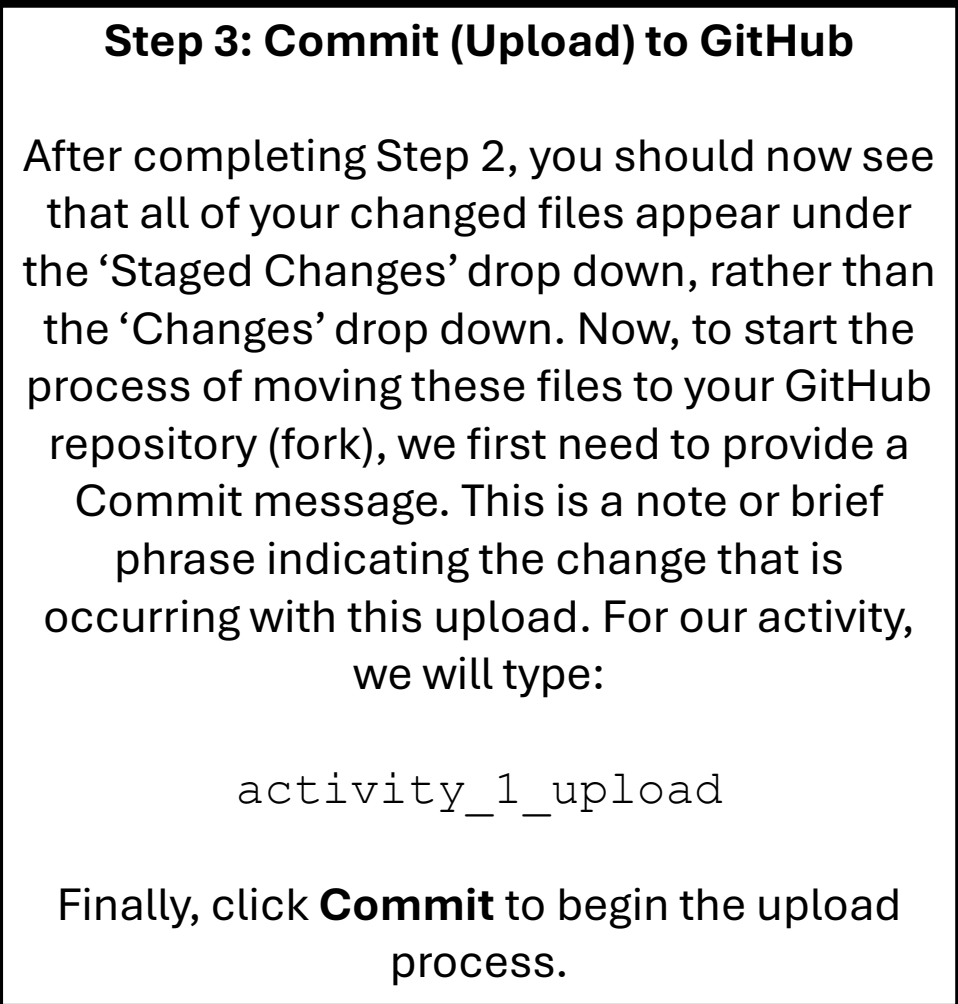
Commit via VS Code





Step 2: Staging the Files

Staging files is how we tell Git which changes (files) to include in our next commit or upload. For this exercise, we will **upload all files**. In VS Code, this is achieved by **clicking the +** next to the 'Changes' drop down (**Hint:** Hover your cursor to see the options).

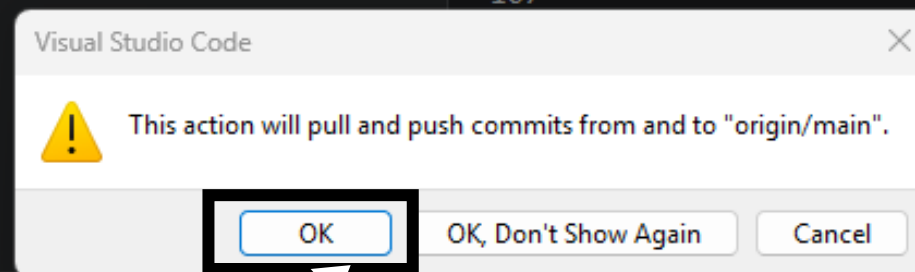


Step 4: Sync Changes

Syncing keeps your local work and the remote repository (GitHub) in agreement. When you click **Sync Changes**, VS Code does two things in one step: it pushes your committed changes up to the remote and pulls down any new changes made directly in GitHub. You'll find it at the bottom status bar or in the Source Control panel, often showing arrows with numbers, ↓ for commits waiting to come down, ↑ for commits waiting to go up. Committing saves your work locally; syncing is what shares it and brings you up to date.

To sync your new changes, click **Sync Changes**.

The screenshot shows the VS Code interface with the Source Control panel on the left. The 'CHANGES' section has a message input field and a 'Sync Changes 1 ↑' button. The 'GRAPH' section shows a commit history. The main editor shows an R script file named 'gam_flusight_forecast.R' with code for a forecasting model. The status bar at the bottom indicates the current file is 'main' and shows the number of changes (1) and the status of the repository (0 changes, 2 files, 26 lines).



Step 5: Confirming Upload to the Main Branch

The main branch is your repository's **official, up-to-date version of your project**. It's what people see by default and what other branches typically merge back into once their changes are ready. Think of it as the trunk: experimental work happens on separate branches, but main holds the version you consider stable and current.

Click OK.

Note: You may be prompted to sign into GitHub after this step.

The screenshot shows a GitHub repository page for 'SISMID-2026', which is a fork of 'bleicham/SISMID-2026'. The repository is public and has 0 stars, 0 forks, and 0 watchers. The main branch is 'main', and there is 1 branch and 0 tags. The repository is currently 1 commit ahead of and 1 commit behind the upstream repository. The commit history shows a recent commit by 'bleicham' with the message 'activity_1_upload' 2 minutes ago. The file list shows a folder 'USE-ME/Bleichrodt_Amanda' and a file 'README.md'. The README file content is visible, showing the title 'SISMID 2026: A Human-AI Teaming Approach to Infectious Disease Modeling'. A black box highlights the 'USE-ME/Bleichrodt_Amanda' folder and the 'activity_1_upload' commit, with an arrow pointing to the commit message.

bleicham-tester / SISMID-2026

SISMID-2026 Public

forked from bleicham/SISMID-2026

main 1 Branch 0 Tags

Go to file Add file <> Code

This branch is 1 commit ahead of and 1 commit behind bleicham/SISMID-2026:main . Contribute Sync fork

bleicham activity_1_upload 1345e0c · 2 minutes ago 27 Commits

USE-ME/Bleichrodt_Amanda activity_1_upload 2 minutes ago

README Update README for improved clarity and structure 3 hours ago

SISMID 2026: A Human-AI Teaming Approach to Infectious Disease Modeling

About No description, website, or topics provided.

Readme Activity 0 stars 0 watching 0 forks

Releases No releases published Create a new release

Packages No packages published Publish your first package

Step 6: Back to our Fork

Now, let's double-check everything worked. First, **navigate** back to your forked repo and **refresh** the page. You should now see the newly committed (uploaded) folder within the *USE-ME* directory. As we continue through the lab, we will add more output to this folder.

Commit via GitHub

***Does not require Git download**

bleicham-tester / SISMID-2026

Code

Pull requests

Actions

Projects

Wiki

Security and quality

Insights

Settings

Type to search

SISMID-2026

Public

forked from bleicham/SISMID-2026

Pin

Watch 0

Fork 0

Star 0

main

1 Branch

0 Tags

Go to file

Add file

Code

This branch is 2 commits ahead of and 1 commit behind bleicham/SISMID-2026:main

Contribute

Sync fork

bleicham-tester

Delete USE-ME/Bleichrodt_Amanda directory

72f4290 · now

37 Commits

Lab Materials/Day 1 - Lab Materials/Activity 1

lab-materials

14 hours ago

README.md

Update README for improved clarity and structure

19 hours ago

README

SISMID 2026: A Human-AI Teaming Approach to Infectious Disease Modeling

Building an Influenza Forecasting Pipeline · July 22 through July 24, 2026

1. Overview

Every week during influenza season, public health decision-makers need to anticipate what is coming: how many people will be hospitalized next week, when the peak will hit, and how high it will go. In this lab you work as a human-AI team to build a national influenza hospitalization forecasting system from real CDC data.

You define the vision and the rules. AI helps build and implement the solution. Think of AI as a very capable intern: fast and helpful, but it still needs direction. The clearer the guidance, the better the results. At the end of the day, you are still the expert making the decisions.

About

No description, website, or topics provided.

Readme

Activity

0 stars

0 watching

0 forks

Releases

Step 1: Navigate to your GitHub Fork

We can also upload or commit our results via GitHub directly. This circumvents the need for the Git download we discussed earlier but takes a bit more time to upload files. First, you will want to **navigate** to your GitHub fork.

The screenshot shows the GitHub interface for a repository named 'SISMID-2026', which is a fork of 'bleicham/SISMID-2026'. The repository is public and has 1 branch and 0 tags. The main branch is 'main'. The repository is 2 commits ahead and 1 commit behind the upstream repository. The repository contains a directory 'Lab Materials/Day 1 - Lab Materials/Activity 1' and a file 'README.md'. The repository has 37 commits and 0 forks. The repository is currently being viewed on the 'main' branch. The repository is currently being viewed on the 'main' branch. The repository is currently being viewed on the 'main' branch.

Annotations:

- A black box highlights the 'Add file' button in the upper right-hand corner.
- A black box highlights the 'Upload files' option in the dropdown menu.
- Two black arrows point from the text box below to the 'Add file' button and the 'Upload files' option.

Step 2: Starting the Process of Uploading Files

To start the process of uploading our files, we first are going to click the **Add file** button in the upper right-hand corner.

As we already have the files on our local machine we want to upload to GitHub, we need to click **Upload files** rather than Create new file. This tells GitHub that we want to search our local machine rather than create something directly in GitHub.

About

No description, website, or topics provided.

Releases

No releases published
[Create a new release](#)

Packages

No packages published
[Publish your first package](#)

Contributors

No contributors

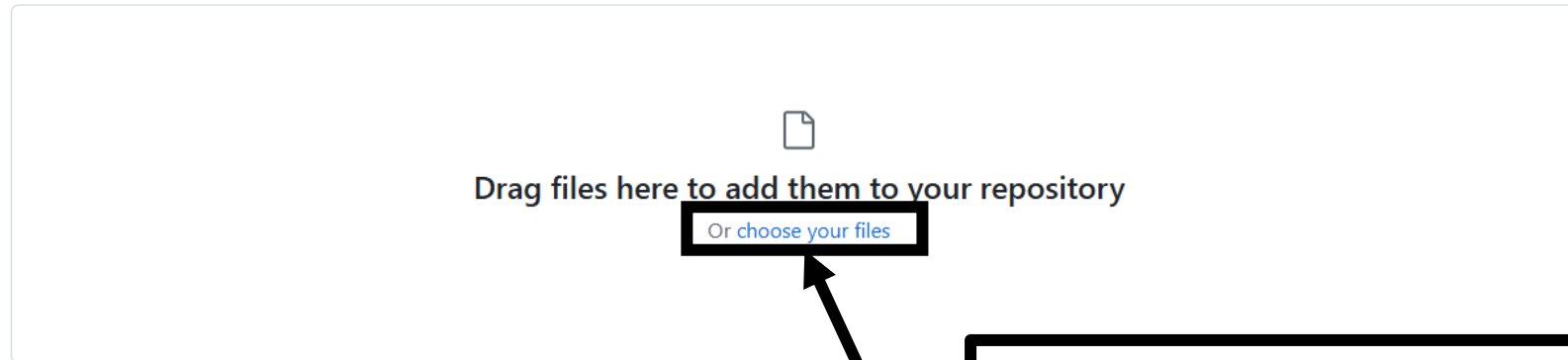
Languages

HTML 99.2% R 0.8%

Suggested workflows

Based on your tech stack

SISMID-2026 /



Commit changes

Add files via upload

Add an optional extended description...

☒ Commit directly to the `main` branch.

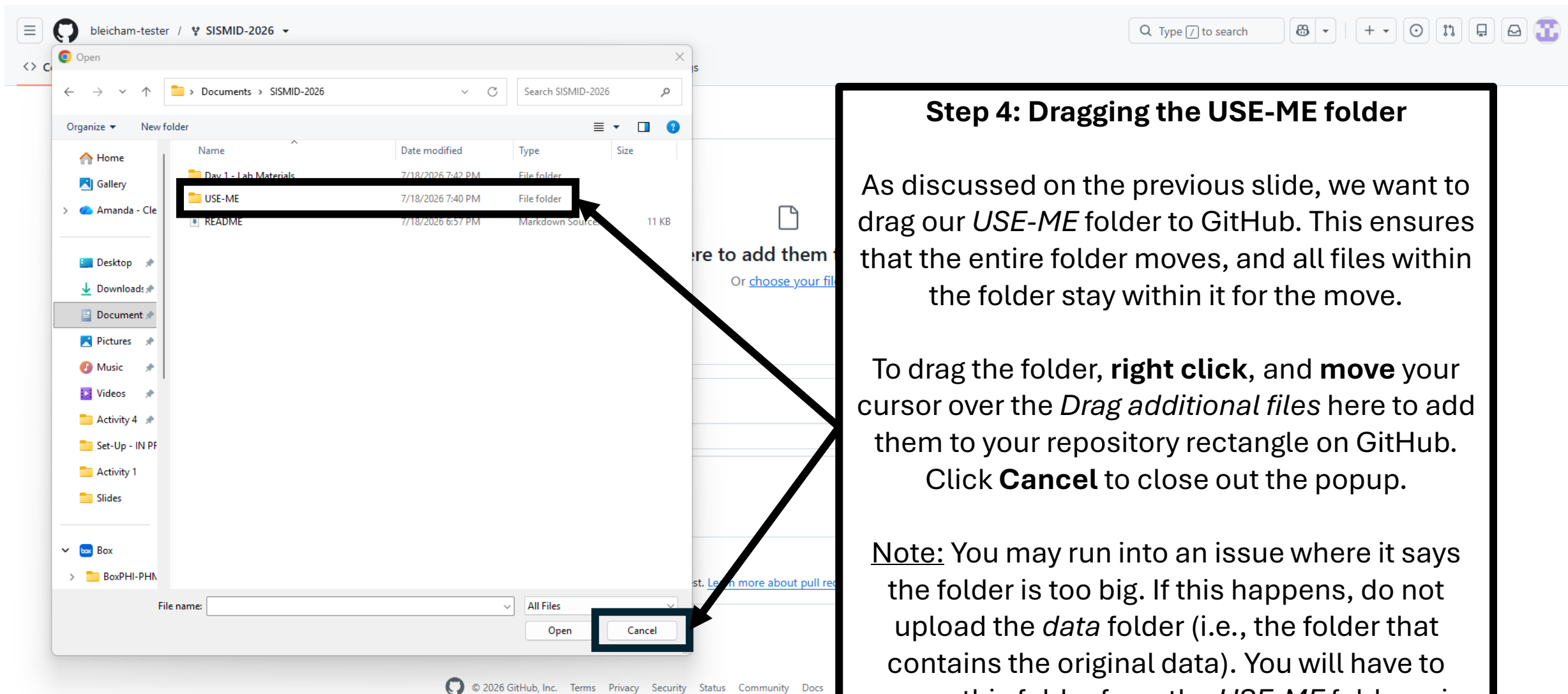
☐ Create a new branch for this commit and start a pull request. [Learn more about pull requests.](#)

Commit changes

Cancel

Step 3: Uploading the Files

To upload files, we are going to first click **Choose your files**. This will open our file explorer and allow us to drag over our entire *USE-ME* folder. GitHub does not allow you to select an entire folder to upload, only individual files. It will however let you drag over an entire folder, which is what we will do for this activity.



Step 4: Dragging the USE-ME folder

As discussed on the previous slide, we want to drag our *USE-ME* folder to GitHub. This ensures that the entire folder moves, and all files within the folder stay within it for the move.

To drag the folder, **right click**, and **move** your cursor over the *Drag additional files* here to add them to your repository rectangle on GitHub. Click **Cancel** to close out the popup.

Note: You may run into an issue where it says the folder is too big. If this happens, do not upload the *data* folder (i.e., the folder that contains the original data). You will have to remove this folder from the *USE-ME* folder prior to dragging into GitHub.



Drag additional files here to add

Or [choose your files](#)

Step 5: Checking the Uploads

After dragging over the folder, you should see each individual component included within, as well as their respective file paths. If you notice any issues, please let us know and we can help debug!

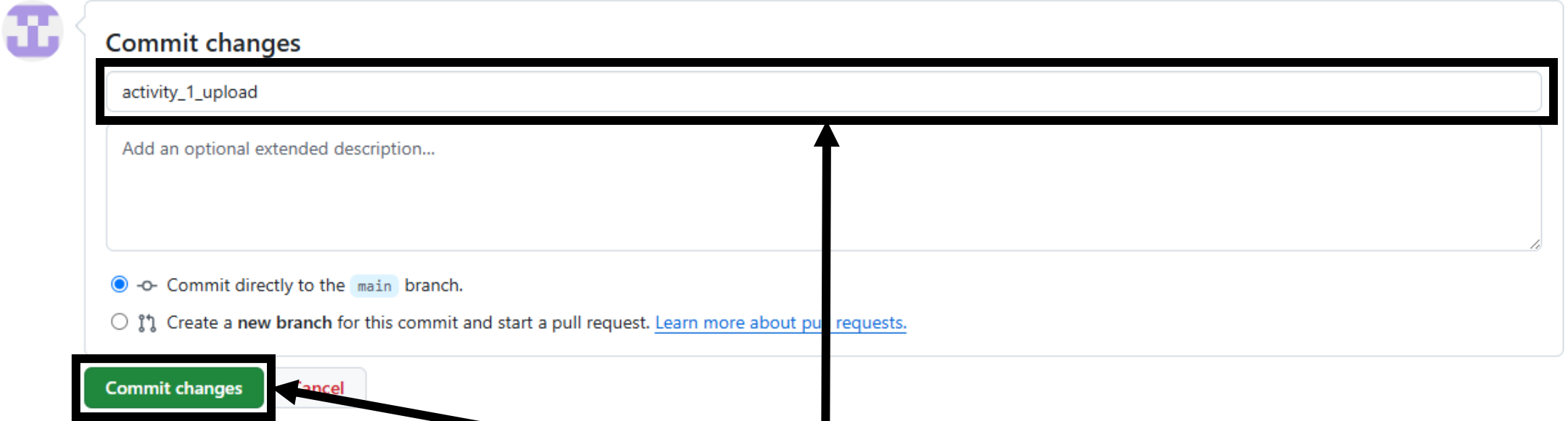
 /USE-ME/Bleichrodt_Amanda/AGENTS.md

 /USE-ME/Bleichrodt_Amanda/output/data/01_cleaning/cleaned_flu_admissions.csv

 /USE-ME/Bleichrodt_Amanda/output/figures/01_cleaning/epicurve_us_flu_admissions.png

 /USE-ME/Bleichrodt_Amanda/rules.md

 /USE-ME/Bleichrodt_Amanda/scripts/01_cleaning.R



The image shows the GitHub 'Commit changes' dialog. A purple GitHub logo is in the top left. The title 'Commit changes' is at the top. Below it is a text input field containing 'activity_1_upload', which is highlighted with a thick black border. Underneath is a larger text area with the placeholder 'Add an optional extended description...'. Below that are two radio button options: the first is selected and says 'Commit directly to the main branch.', and the second is unselected and says 'Create a new branch for this commit and start a pull request. Learn more about pull requests.' At the bottom left is a green 'Commit changes' button, also highlighted with a thick black border. To its right is a grey 'Cancel' button. Two black arrows originate from a point below the text box: one points vertically up to the commit message input field, and the other points diagonally down-left to the green 'Commit changes' button.

Commit changes

activity_1_upload

Add an optional extended description...

☒ Commit directly to the `main` branch.

☐ Create a new branch for this commit and start a pull request. [Learn more about pull requests.](#)

Commit changes Cancel

Step 6: Commit Changes

We are now ready to save our files to GitHub. First, GitHub asks us to give the change a short title, called a commit message. Delete the default text in the top box (it reads *Add files via upload*) and **type** `activity_1_upload` instead. Then **click** the green *Commit changes* button. From now on, always write clear commit messages, so you can easily tell what each change did later.

The screenshot shows a GitHub repository page for 'SISMID-2026' (Public), forked from 'bleicham/SISMID-2026'. The repository has 1 branch (main) and 0 tags. The commit history shows a recent commit by 'bleicham' (activity_1_upload) 2 minutes ago. A file named 'USE-ME/Bleichrodt_Amanda' is highlighted with a black box, and an arrow points to it from the text box below. The README file is also visible, titled 'SISMID 2026: A Human-AI Teaming Approach to Infectious Disease Modeling'.

Step 7: Back to our Fork

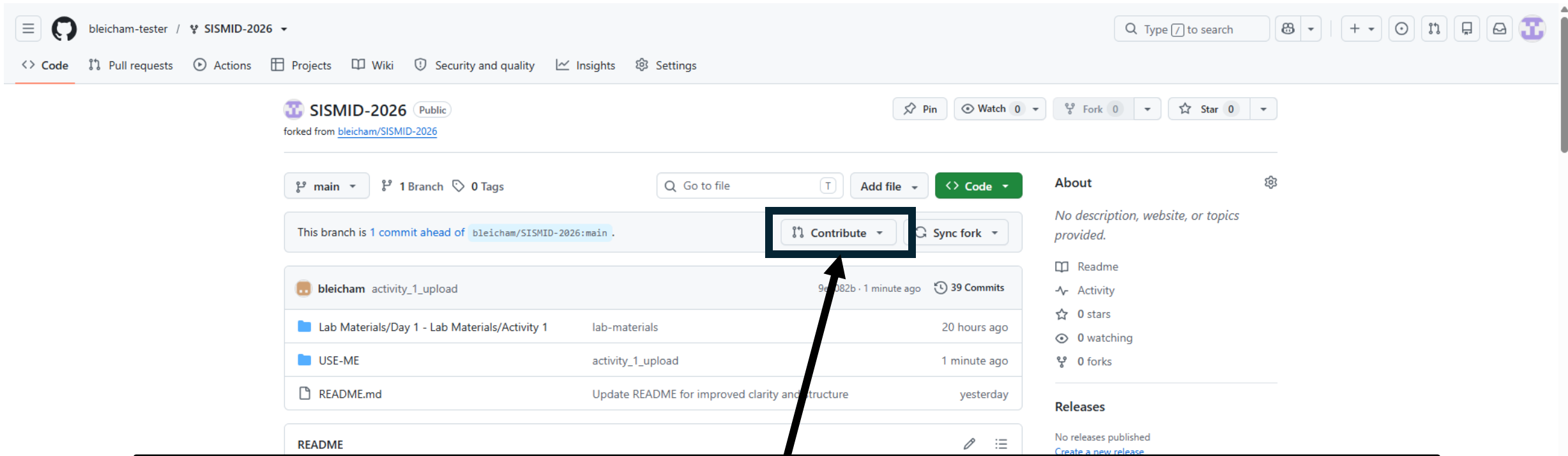
Now, let's double-check everything worked. First, **navigate** back to your forked repo and **refresh** the page. You should now see the newly committed (uploaded) folder within the *USE-ME* directory. As we continue through the lab, we will add more output to this folder.

Please note: When you update GitHub, you only need to upload the files that you changed. If a file did not change, do not re-upload it. This keeps you under GitHub's web upload limits, which only apply when you use the web portal.



Task 2: Submitting a Pull Request

(Merging your fork and the class repository)



Step 1: Starting the *Pull Request* process

Now that we have updated our fork, we need to combine our updates with the main course repository. The goal of this is to get some additional practice with GitHub, including submitting a *Pull Request*. As a reminder, a *Pull Request* is a way of asking the owner of the main project to review the changes you made in your fork and add them to the original repository. In other words, you are saying, “Here is what I changed. Please pull it into the main project.”

Click *Contribute* to start this process.

bleicham-tester / SISMID-2026

Search

CodePull requestsActionsProjectsWikiSecurity and qualityInsightsSettings

SISMID-2026Publicforked from bleicham/SISMID-2026

PinWatch 0Fork 0Star 0

main1 Branch0 Tags

Go to file

Add fileCode

This branch is 1 commit ahead of bleicham/SISMID-2026:main

ContributeSync fork

bleicham activity_1_upload

Lab Materials/Day 1 - Lab Materials/Act20 hours ago39 Commits

USE-ME1 minute ago

README.mdUpdate README for improved clarity and structureyesterday

README

SISMID 2026: A Human-AI Teaming Approach to

About

No description, website, or topics provided.

ReadmeActivity0 stars0 watching0 forks

Releases

No releases publishedCreate a new release

Packages

Languages

HTML 99.3%R 0.7%

Suggested workflows

Based on your tech stack

Step 2: Open the Pull Request

Next, we need to formally open the pull request. To do so, click Open pull request.

Comparing changes

Choose two branches to see what's changed or to start a new pull request. If you need to, you can also [compare across forks](#) or [learn more about diff comparisons](#).

 base repository: bleicham/SISMID-2026 base: main ... head repository: bleicham-tester/SISMID-2026 compare: main

✓ **Able to merge.** These branches can be automatically merged.



Add a title *

Helpful resources

[GitHub Community Guidelines](#)

Add a description

Write

Preview

H

B

I

Add your description here...

 Markdown is supported

 Paste, drop, or click to add files

☒ Allow edits by maintainers

Create pull request

 Remember, contributions to this repository should follow our [GitHub Community Guidelines](#).

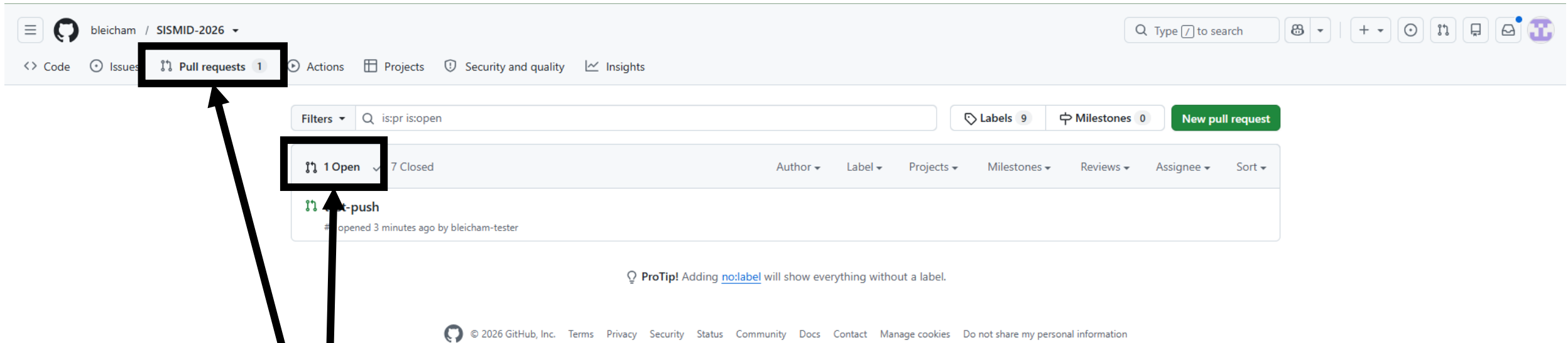
 1 commit

 6 files changed

 1 contributor

Step 3: Comparing changes

We do not need to do much on this page (everything should auto-populate). To finish submitting the pull request, **click** *Create pull request*.



Step 4: Viewing the Pull Request

Our Pull Request has now been created. To view it, open the *Pull requests* tab at the top of the course repository. Any open requests are listed here, so ours, *test-push*, shows up as *1 Open*. Click its title to see the details, including the changes we submitted and any comments from the reviewer.

bleicham / SISMID-2026

CodeIssuesPull requests1AgentsActionsProjectsWikiSecurity and qualityInsightsSettings

SISMID-2026PublicPinWatch0Fork2Star0

main2 Branches0 Tags

Go to fileTAdd fileCode

bleicham Merge pull request #6 from bleicham-tester/main695e2a8 · now40 Commits

Lab Materials/Day 1 - Lab MaterialsActivity 1lab-materials20 hours ago

USE-MEactivity_1_upload3 minutes ago

README.mdUpdate README for improved clarity and structureyesterday

README

SISMID 2026: A Human-AI Teaming Approach to Infectious Disease Modeling

About

No description, website, or topics provided.

ReadmeActivity0 stars0 watching2 forks

Releases

No releases published

Create a new release

you define the vision and the rules. AI helps build and implement the solution. Think of AI as a very capable intern. Fast and helpful, but it still needs direction. The clearer the guidance, the better the results. At the end of the day, you are still the expert making the decisions.

The heart of the lab is a single document, `rules.md`, that you author in plain language. It is the blueprint that describes what the pipeline should do. Across the activities it grows from a data-cleaning spec into a full forecasting

HTML 99.3%R 0.7%

Suggested workflows

Based on your tech stack

Step 5: Viewing the Merged Pull Request

Our Pull Request has now been merged, which means the changes from our fork are now part of the main repository. You can see this at the top of the file list, where the latest commit reads *Merge pull request #6*. The files we uploaded now appear in the repository.

Important Reminder

Every time you **commit** changes to your fork, remember to also submit a **pull request** to the course repository. This keeps your updates merged into the class repo so everyone stays in sync.